

Welding Defect Object Detection

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Problem Statement & Objective

Problem:

- Welding defects affect structural integrity and safety
- Manual inspection is time-consuming and error-prone

Objective:

- Develop a CNN-based object detector
- Identify and localize welding defects in images and video
- Achieve reliable detection with real-time inference capability

Dataset

Dataset

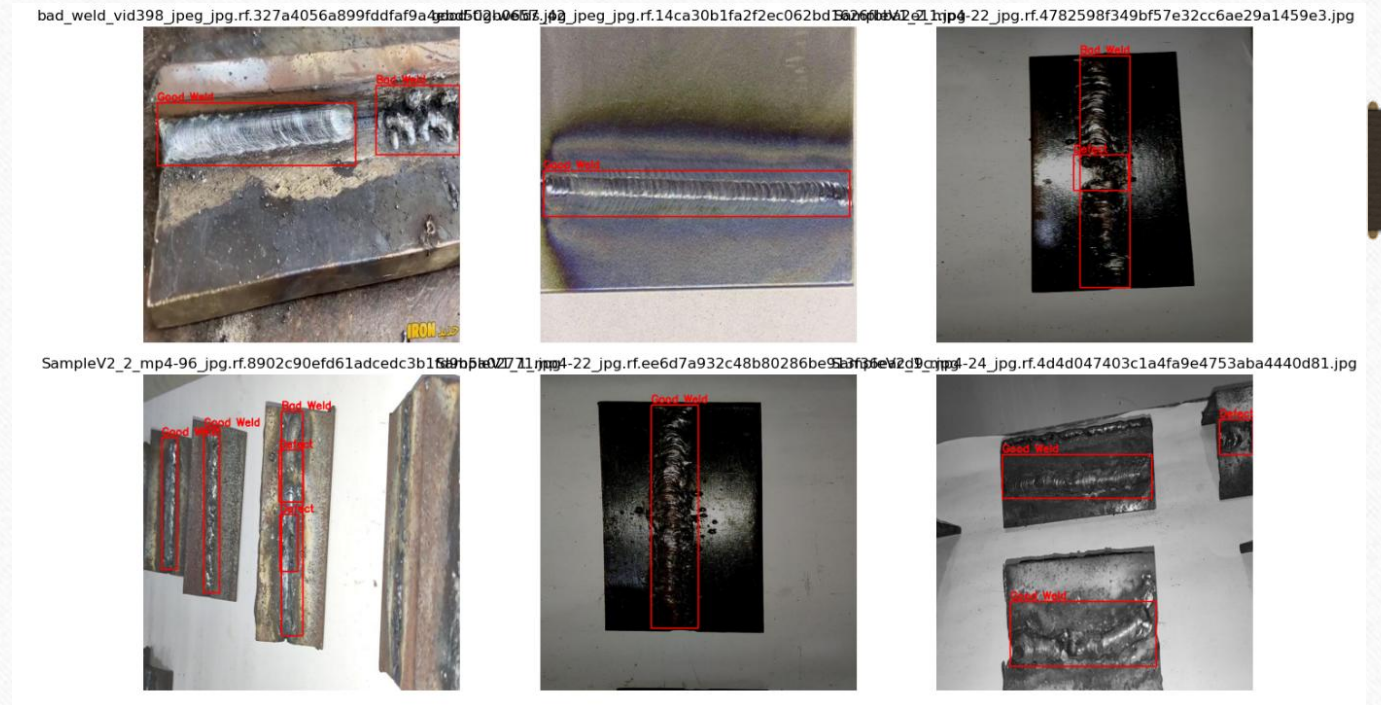
- Welding Defect Object Detection (Kaggle)
- Annotated in YOLO format

Classes

- Bad Weld
- Good Weld
- Defect

Data Split

- Train: 70%
- Validation: 15%
- Test: 15%



Data Preparation & Verification

Steps:

- Dataset inspection and integrity check
- Verification of images and labels
- Visualization of bounding boxes
- Fixed random seed for reproducibility

Why important?

- Ensures correct annotations
- Prevents training on corrupted data

Model Development

Architecture:

- YOLOv8 (YOLOv8s and YOLOv8n)

Framework:

- PyTorch (Ultralytics)

Training Setup:

- Transfer learning using pretrained COCO weights
- Input size: 640×640
- Epochs: 50

Why YOLOv8?

- Fast, accurate, real-time capable
- End-to-end detection in a single forward pass

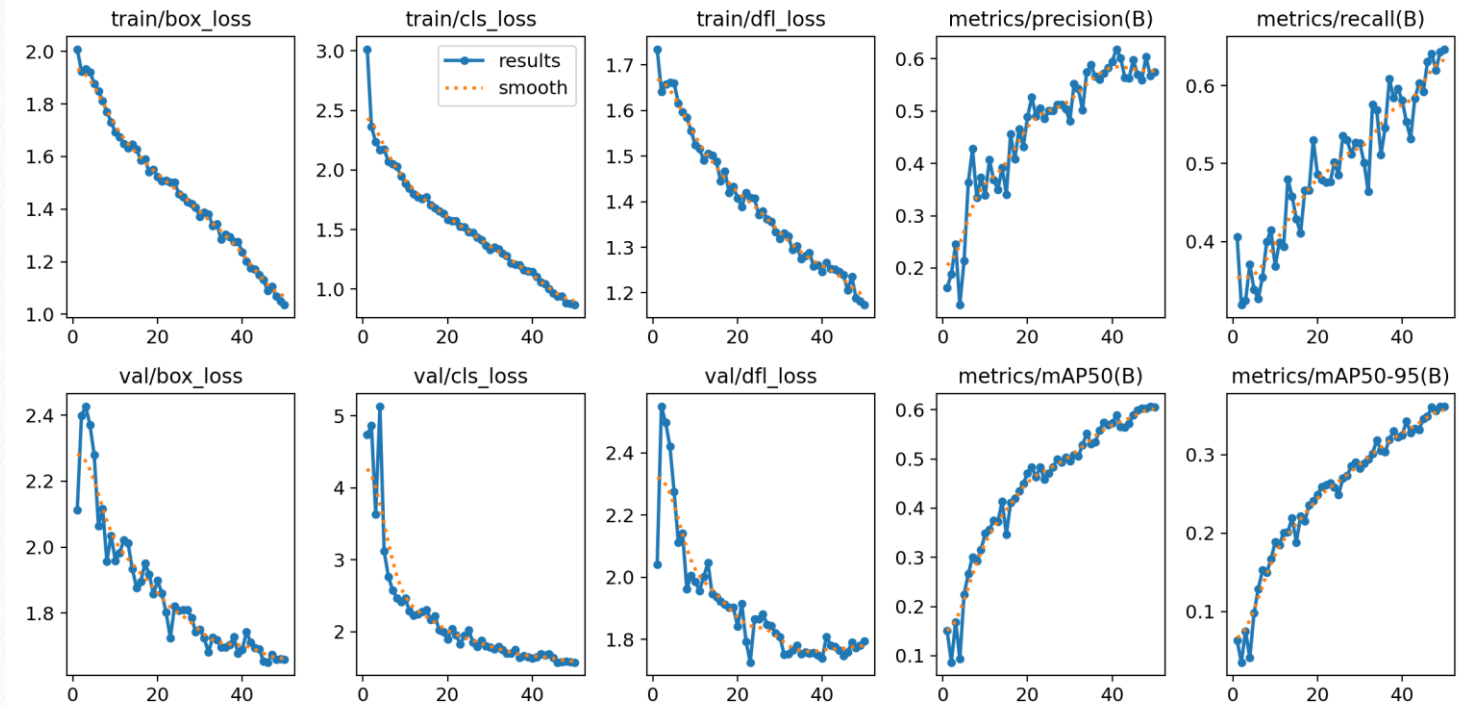
Training Results

Metrics Tracked:

- Box loss
- Classification loss
- mAP@0.5

Observation:

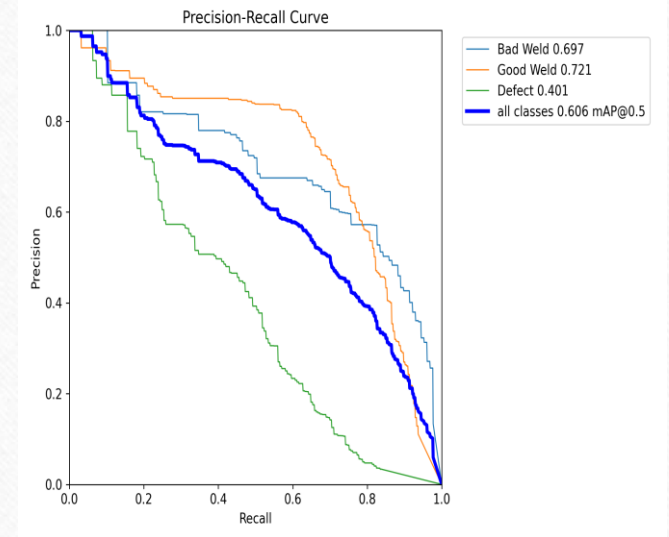
- Loss decreases steadily
- mAP improves across epochs



Model Evaluation

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.9 5
YOLOv8n	0.609	0.511	0.567	0.317
YOLOv8s	0.631	0.553	0.591	0.333

- YOLOv8s performs better due to higher model capacity



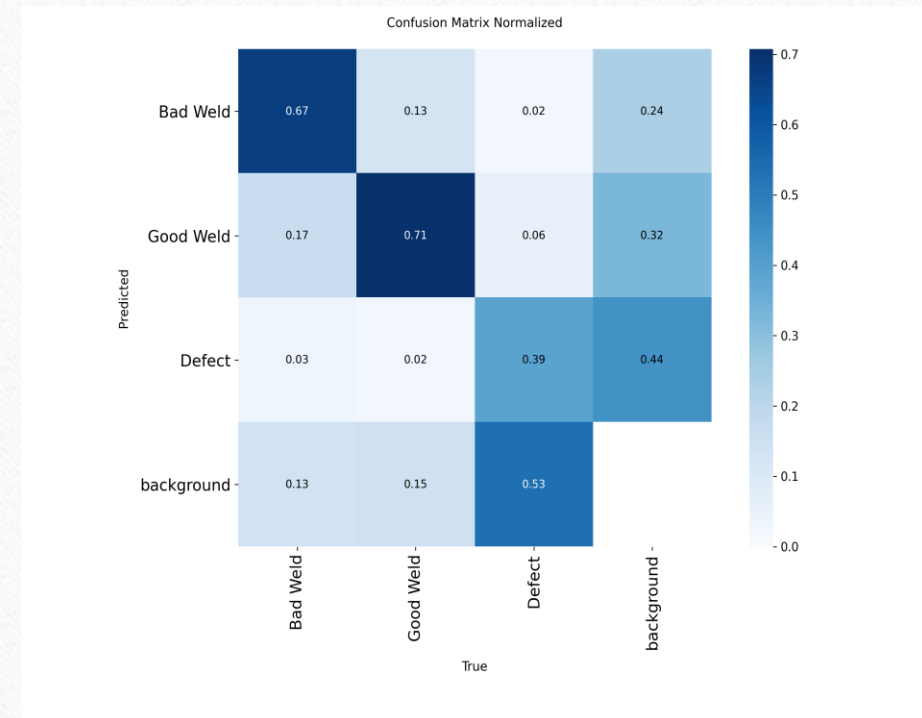
Qualitative Results & Failure Analysis

Good Detections:

- Accurate localization of Bad Weld and Good Weld
- Multiple objects detected per image

Failure Cases:

- Small or low-contrast defects
- Defect class confused with weld texture



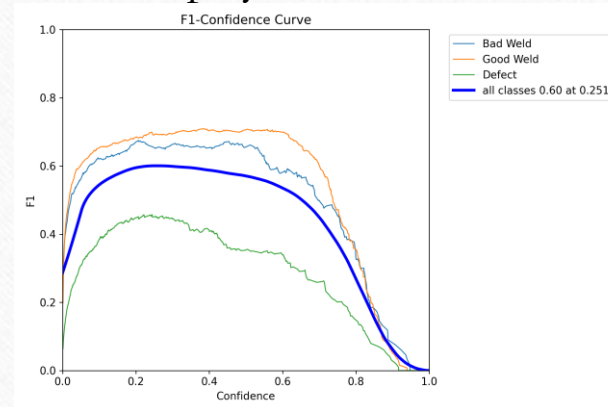
Video Inference & FPS

Video Inference:

- Detector applied to a short welding clip
- Bounding boxes and class labels displayed

Performance:

- Frames: 74
- Average FPS: **17.68 FPS**



Conclusion

- Successfully trained a CNN-based welding defect detector
- Achieved good accuracy and real-time performance
- YOLOv8s performed best among tested models